

SEMESTER END EXAMINATIONS – JULY 2024

Program	: B.E. – Information Science and Engineering	Semester	: VI
Course Name	: Machine Learning	Max. Marks	: 100
Course Code	: IS62	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- Define Machine learning? Explain each types of learning with an example. CO1 (10)
 - What is Curse of Dimensionality? Discuss the various problems raised due to curse of dimensionality. CO1 (06)
 - Explain with an example each the difference between the following: CO1 (04)
 - Bias and Variance of the ML Model
 - Variance and Covariance of the ML Model.
- Briefly explain the process by which machine learning algorithms can be selected, applied, and evaluated for the problem. CO1 (06)
 - Differentiate between the following: CO1 (04)
 - Over-fitting and Under-fitting
 - Classification and Regression ML algorithms.
 - In a brain MRI image Classification Model to separate 5000 brain MRI samples were examined. The number of TP = 800, FP = 500, TN = 3500, FN = 200. Do the following for this brain MRI image Classification Model: CO1 (10)
 - Draw the Confusion Matrix
 - Calculate the Accuracy value
 - Assess the Sensitivity Value
 - Estimate the Specificity Value
 - Measure the Error Rate
 - Determine the Precision Value
 - Compute the Recall Value
 - Evaluate the F-Measure
 - Is the data sample given to you a Balanced or an Imbalanced one? Justify your answer.
 - Is this Model a 'High Precision' or 'High Recall' model? Justify your answer.

UNIT - II

- Calculate the dissimilarity matrix for attributes of mixed types. CO2 (08)

RollNo	Marks	Grade
1	80	A
2	70	B
3	92	B
4	94	A

- Discuss the Properties of Principal Component Analysis. And also highlight the steps to calculate PCA. CO2 (06)

- c) Classify the following attributes as binary, discrete, or continuous. Also classify them as qualitative (nominal or ordinal) or quantitative (interval or ratio). CO2 (06)
- Time in terms of AM or PM.
 - Brightness as measured by a light meter.
 - Brightness as measured by people's judgments.
 - Angles as measured in degrees between 0° and 360° .
4. a) Given two objects represented by the tuples (22, 1, 42, 10) and (20, 0, 36, 8): CO2 (08)
- Compute the Euclidean distance between the two objects.
 - Compute the Manhattan distance between the two objects.
 - Compute the Minkowski distance between the two objects, using $q = 3$.
 - Compute the supremum distance between the two objects.
- b) Briefly outline how to compute the dissimilarity between objects described by the following: CO2 (06)
- Nominal attributes
 - Asymmetric binary attributes.
- c) Illustrate the Linear Discriminant Analysis with an example. Suggest any one application where LDA can be used. CO2 (06)

UNIT - III

5. a) The sales of a company (in million dollars) for each year are shown in the table below. CO3 (06)
- | X (Years) | 2005 | 2006 | 2007 | 2008 | 2009 |
|-----------|------|------|------|------|------|
| Y (Sales) | 12 | 19 | 26 | 37 | 45 |
- Find the least square regression line $y = a x + b$
 - Use the least square regression line as a model to estimate the sales of the company in 2012.
- b) Identify the first splitting attribute for decision tree by using ID3 algorithm with the following dataset. CO3 (06)

Major	Experience	Tie	Hired?
CS	programming	pretty	NO
CS	programming	pretty	NO
CS	management	pretty	YES
CS	management	ugly	YES
business	programming	pretty	YES
business	programming	ugly	YES
business	management	pretty	NO
business	management	pretty	NO

- c) Explain K-nearest neighbor algorithm with an example. CO3 (08)
6. a) Write and explain the classification tree algorithm. Describe the concept of "Gini Index" as an attribute selection method. CO3 (08)
- b) Apply Naive Bayes Classifier to the dataset and determine if the red, domestic, SUV is stolen? CO3 (06)

Example No.	Color	Type	Origin	Stolen?
1	Red	Sports	Domestic	Yes
2	Red	Sports	Domestic	No
3	Red	Sports	Domestic	Yes
4	Yellow	Sports	Domestic	No
5	Yellow	Sports	Imported	Yes
6	Yellow	SUV	Imported	No
7	Yellow	SUV	Imported	Yes
8	Yellow	SUV	Domestic	No
9	Red	SUV	Imported	No
10	Red	Sports	Imported	Yes

- c) Suppose that the following are a set of points in two classes: CO3 (06)
 Class 1: $\{(1,1), (1,2), (2,1)\}$
 Class 2: $\{(0,0), (1,0), (0,1)\}$
 Plot them and find the optimal separating line. What are the support vectors, and what is the margin?

UNIT- IV

7. a) Write k-means clustering algorithm. Apply k-means clustering algorithm for the following datasets to create two clusters. Tabulate all the assignments. CO4 (07)

Sample No.	X	Y	Assignment
1	85	52	
2	70	36	
3	68	30	
4	79	38	
5	82	42	
6	88	47	

- b) Differentiate between 'AGNES' and 'DIANA' as Hierarchical Clustering techniques. CO4 (06)
 c) Use the similarity matrix in the following table to perform single and complete link hierarchical clustering. Show your results by drawing a dendrogram. The dendrogram should clearly show the order in which the points are merged. CO4 (07)

	P1	P2	P3	P4	P5
P1	1.00	0.10	0.41	0.55	0.35
P2	0.10	1.00	0.64	0.47	0.98
P3	0.41	0.64	1.00	0.44	0.85
P4	0.55	0.47	0.44	1.00	0.76
P5	0.35	0.98	0.85	0.76	1.00

8. a) Explain how Elbow method is used for choosing the right value of k in K-means clustering. CO4 (06)
 b) Cluster the following 8 data points A1(2,10), A2(2,5), A3(8,4), A4(5,8), A5(7,5), A6(6,4), A7(1,2), A8(4,9). Use k-means clustering, k = 3. Initial centroids are data points A1, A4 and A7. CO4 (08)
 c) What are the similarities and differences between average-link clustering and k-means? CO4 (06)

UNIT - V

9. a) Explain the guidelines need to be considered before we start experimentation in machine learning. CO5 (10)
 b) With an example explain the following: CO5 (10)
 i) K-fold cross validation ii) Bootstrapping.
10. a) Explain the basic principles need to be followed during experimental design. CO5 (08)
 b) illustrate how process generates an output given an input and is affected by controllable and uncontrollable factors. CO5 (12)
